STAFF REVIEW · H446 201 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Because traversal is permitted both ways.

2. Left.

3. 0, 1, 2, 3, 4.

4. 5, since 17 = 2×6 + 5.

I Do — source worked steps

1. Predict and explain — Insert 6 at index 1, because 6 MOD 5 = 1.

2. Check and extend — 11 collides at 1, so store it at 2.

3. Check and extend — 16 collides at 1 and 2, so store it at 3.

We Do — guided CHECK

1. Predict and explain
8 goes into bucket 3.
Check and extend
13 collides at 3 and probes to 4.
18 collides at 3 and 4; linear probing wraps around to 0, which is empty, so 18 goes into bucket 0.

You Do — source mark schemes

1. 24 MOD 7 = 3.

2. All hash to 4. Insert 4 at index 4; 9 probes to 0; 14 probes past occupied 4 and 0 to index 1.

3. More occupied buckets lead to longer probe sequences, clustering and slower expected lookup and insertion.

Exit Ticket — source answer

1. True. A collision requires a resolution strategy such as linear probing.